

SQL PRACTICE QUESTIONS

1. Database and Table Creation

- Q1. Create a database named COMPANY_DB.
 - Q2. Create an EMPLOYEES table with appropriate columns and constraints.
 - Q3. Insert at least 10 employee records.
 - Q4. Display all employee records.
 - Q5. Display only employee names.
 - Q6. Display employee name and salary.
 - Q7. Find distinct departments.
 - Q8. Find distinct cities.
 - Q9. Display the structure of the EMPLOYEES table.
 - Q10. Drop the EMPLOYEES table.
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2. Constraints

- Q11. Create a STUDENTS table using PRIMARY KEY, NOT NULL, UNIQUE, CHECK, and DEFAULT constraints.
- Q12. Insert valid records into the STUDENTS table.
- Q13. Try inserting a duplicate email value.
- Q14. Try inserting NULL into a NOT NULL column.
- Q15. Try inserting an age less than 18.
- Q16. Insert a record without specifying the city.
- Q17. Add a UNIQUE constraint to a new column.
- Q18. Drop a column having a UNIQUE constraint.
- Q19. Add a CHECK constraint on marks.
- Q20. Add a JOINING_DATE column with a default value.
- Q21. Modify a nullable column to NOT NULL.
- Q22. Display all constraints on the table.
- Q23. Explain the difference between PRIMARY KEY and UNIQUE KEY.
- Q24. Create a SUBJECT table.
- Q25. Create a TEACHER table with a FOREIGN KEY.
- Q26. Insert records into the SUBJECT table.

- Q27. Insert records into the TEACHER table.
- Q28. Try inserting a teacher with a non-existing subject ID.
- Q29. Display teachers with their subject names.
- Q30. Explain the concept of a FOREIGN KEY.
- Q31. Create DEPARTMENT and EMPLOYEE tables using a FOREIGN KEY.
- Q32. Count employees department-wise.
- Q33. Find the department with the maximum employees.
- Q34. Find records with missing parent references.
- Q35. Delete a parent record referenced by a child table.
- Q36. Create a FOREIGN KEY using ON DELETE CASCADE.
- Q37. Test ON DELETE CASCADE.
- Q38. Create a FOREIGN KEY using ON DELETE SET NULL.
- Q39. Test ON DELETE SET NULL.
- Q40. Find departments with no employees.
- Q41. Move an employee from one department to another.
- Q42. Create STUDENT and COURSE tables.
- Q43. Create an ENROLLMENT table for a many-to-many relationship.
- Q44. Explain many-to-many relationships.
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3. Filtering Data

- Q45. Find employees with salary greater than 50,000.
- Q46. Find employees from Pune.
- Q47. Find employees not from Mumbai.
- Q48. Find employees with salary between 30,000 and 60,000.
- Q49. Find employees working in HR.
- Q50. Find employees in HR or IT.
- Q51. Find employees with salary less than 40,000.
- Q52. Find employees who joined after 2023.
- Q53. Find employees from Pune earning more than 50,000.
- Q54. Find employees not working in Sales.
- Q55. Use the LIKE operator to filter records.

4. Sorting

- Q56. Sort employees by salary in ascending order.
- Q57. Sort employees by salary in descending order.
- Q58. Sort employees by name.
- Q59. Sort employees by department and salary.
- Q60. Display the top 5 highest-paid employees.

5. String Functions

- Q61. Convert employee names to uppercase.
- Q62. Convert employee names to lowercase.
- Q63. Find the length of each employee name.
- Q64. Display the first 3 characters of each employee name.
- Q65. Concatenate employee name and city.
- Q66. Replace Pune with Pune City.
- Q67. Find employees whose names start with 'A'.
- Q68. Find employees whose names end with 'A'.
- Q69. Find employees whose names contain 'SH'.
- Q70. Remove leading and trailing spaces using TRIM().
- Q71. Display employee names using SUBSTRING().

6. Aggregate Functions

- Q72. Find the total salary paid.
- Q73. Find the average salary.
- Q74. Find the highest salary.
- Q75. Find the lowest salary.
- Q76. Count total employees.
- Q77. Count employees department-wise.
- Q78. Find average salary department-wise.
- Q79. Find highest salary department-wise.

Q80. Find lowest salary department-wise.

Q81. Find departments having more than two employees.

7. GROUP BY and HAVING

Q82. Count employees city-wise.

Q83. Find average salary city-wise.

Q84. Find departments whose average salary exceeds 50,000.

Q85. Find cities having more than three employees.

Q86. Find the department with the maximum employees.

Q87. Find total salary department-wise.

Q88. Find total salary city-wise.

Q89. Find departments whose salary sum exceeds 100,000.

Q90. Find cities where average salary exceeds 40,000.

Q91. Find department-wise employee count.

8. Subqueries

Q92. Find employees earning the highest salary.

Q93. Find employees earning the second highest salary.

Q94. Find employees earning the lowest salary.

Q95. Find employees earning above average salary.

Q96. Find employees earning below average salary.

Q97. Find employees working in the same department as RAM.

Q98. Find employees whose salary equals the maximum salary.

Q99. Find employees earning more than all HR employees.

Q100. Find the department having the highest average salary.

9. Joins

Q101. Create a DEPARTMENTS table.

Q 102 INSERT INTO DEPARTMENT TABLE

Q102. Perform an INNER JOIN.

- Q103. Perform a LEFT JOIN.
 - Q104. Perform a RIGHT JOIN.
 - Q105. Perform a FULL JOIN.
 - Q106. Display employee names with department names.
 - Q107. Find employees without departments.
 - Q108. Count employees department-wise using JOIN.
 - Q109. Find the highest-paid employee in each department.
 - Q110. Find departments with no employees.
 - Q111. Find total salary department-wise.
 - Q112. Explain different types of joins.
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10. Self Joins

- Q113. Create a MANAGER_EMPLOYEE table.
 - Q114. Display employee and manager names.
 - Q115. Find employees without managers.
 - Q116. Find managers supervising more than two employees.
 - Q117. Display employee-manager pairs.
 - Q118. Display organizational hierarchy.
 - Q119. Explain Self Join with an example.
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11. Window Functions

- Q120. Rank employees by salary.
- Q121. Dense rank employees by salary.
- Q122. Generate row numbers department-wise.
- Q123. Find the highest-paid employee in each department.
- Q124. Find the second highest-paid employee in each department.
- Q125. Calculate running salary totals.
- Q126. Calculate department-wise cumulative salary.
- Q127. Display previous employee salary using LAG().
- Q128. Display next employee salary using LEAD().
- Q129. Compare salary with the previous employee.

12. CTEs

- Q130. Find employees earning above company average salary using a CTE.
- Q131. Find the highest-paid employee in each department using a CTE.
- Q132. Find the second highest-paid employee in each department using a CTE.
- Q133. Generate department-wise rankings using a CTE.
- Q134. Use multiple CTEs in a single query.

13. Views

- Q135. Create a view showing employees with salary greater than 50,000.
- Q136. Display records from a view.
- Q137. Create a department-wise salary summary view.
- Q138. Create a simple employee view.
- Q139. Update data through a view.
- Q140. Drop a view.
- Q141. Create a view for HR employees.
- Q142. Display records from the HR view.
- Q143. Create a view showing the highest salary.
- Q144. Create a view using a LEFT JOIN.
- Q145. Create a view using multiple tables.
- Q146. Explain the difference between a table and a view.

14. Indexes

- Q147. Create an index on employee name.
- Q148. Create a composite index.
- Q149. Display all indexes on a table.
- Q150. Drop an index.
- Q151. When should indexes be used?
- Q152. When should indexes be avoided?
- Q153. Explain Clustered vs Non-Clustered Indexes.

- Q154. Create an index on an email column.
- Q155. Compare query performance before and after indexing.
- Q156. What are the disadvantages of excessive indexing?

15. Stored Procedures

- Q157. Create a stored procedure to display all employee records.
- Q158. Execute a stored procedure that displays all employees.
- Q159. Create a stored procedure to fetch employee details by employee ID.
- Q160. Create a stored procedure to fetch employees based on salary.
- Q161. Create a stored procedure that returns the total employee count using an OUT parameter.
- Q162. Create a stored procedure to insert a new employee record.
- Q163. Create a stored procedure to update an employee's salary.
- Q164. Create a stored procedure to delete an employee record.
- Q165. Create a stored procedure using an IF condition to check whether an employee exists.
- Q166. Explain the difference between a Stored Procedure and a Function.
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16. Triggers

- Q167. Create a LOG table to store trigger activity.
- Q168. Create an AFTER INSERT trigger on the EMPLOYEES table.
- Q169. Insert a record and verify that the AFTER INSERT trigger executes successfully.
- Q170. Create a BEFORE INSERT trigger to log insert attempts.
- Q171. Create a trigger to prevent insertion of negative salary values.
- Q172. Test the trigger by attempting to insert an employee with a negative salary.
- Q173. Create an AFTER UPDATE trigger to track employee record updates.
- Q174. Create a SALARY_AUDIT table to store salary changes.
- Q175. Create a trigger that records salary changes into the SALARY_AUDIT table.
- Q176. Create a BEFORE DELETE trigger to log employee deletions.
- Q177. Create a DELETED_EMPLOYEES table to store deleted employee records.
- Q178. Create a trigger that stores deleted employee details before deletion.
- Q179. Delete an employee and verify that audit information is captured.
- Q180. Display all trigger-generated logs.

17. Transactions

Q181. Create an ACCOUNTS table.

Q182. Insert sample account records.

Q183. Transfer money from one account to another using a transaction.

Q184. Commit a transaction using COMMIT.

Q185. Demonstrate transaction rollback using ROLLBACK.

Q186. Create a SAVEPOINT within a transaction.

Q187. Roll back a transaction to a SAVEPOINT.

Q188. Simulate a failed transaction and roll back all changes.

Q189. Display account balances after transaction operations.

Q190. Explain the ACID properties of database transactions.

ACID Related Theory Questions

Q191. What is Atomicity? Explain with an example.

Q192. What is Consistency? Explain with an example.

Q193. What is Isolation? Explain with an example.

Q194. What is Durability? Explain with an example.